# Viva Sheet: Time-to-Employment for Sri Lankan IT Graduates

M.S.F. Shazna · Every number below matches the thesis (checked 15 Oct 2026). Speak the answers in your own words. Don't read them.

## 1. 30-second pitch

Sri Lanka produces about 17,000 IT graduates a year, but no graduate can see how long *their* search for a first IT job is likely to take. Existing tools only classify or match jobs, and they throw away graduates who are still searching. I built an explainable **survival-analysis** system, trained on **885 real graduates** and triangulated with **239 employers** and **324 real job ads**. It predicts a **time range**, explains what is helping or holding a profile back, and shows skill gaps against real demand. On a test set used only once, it ranked graduates better than chance and better than ChatGPT (**0.619 vs 0.539**), and users rated it usable (**SUS 76.3**).

## 2. Numbers to memorise

| What | Number |
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| Responses collected | 1,175 graduates · 245 employers (239 analysed) · 324 job ads from 211 employers (12 Jul – 11 Oct 2026) |
| Modelling graduates and split | 885 → train 620 / validation 133 / test 132 (70/15/15, nobody in two parts); 230 final-year students set aside (no outcome yet) |
| Still searching (censored) | About a quarter (24.2–24.8% in each split) |
| Found an IT job by 3 / 6 / 12 / 24 months | 43.6% / 67.0% / 80.9% / 89.8% · median 4 months |
| Sample-size target | Cochran: n = 384 (Z = 1.96, p = 0.5, e = 0.05) → target about 400 |
| Models compared | Cox PH, Elastic Net Cox, AFT (Weibull, log-normal, log-logistic), DeepSurv |
| Selected model | AFT Log-Normal · validation Uno's C 0.693 (0.636–0.745) |
| **Test result (used once)** | **Uno's C 0.619 (95% CI 0.557–0.683)** · Harrell's C 0.626 · IBS 0.167 (Kaplan–Meier 0.164) |
| ChatGPT on the same 132 graduates | Uno's C 0.539 · difference 0.081 (95% CI 0.030–0.131) |
| Strongest factors | GPA and internship: 1 SD more → about 20% and 16% shorter search |
| Skill gap | Graduates below employer importance on 25 of 26 skills (cloud, testing, DevOps largest) |
| Employers' ratings vs ads | Only moderate agreement (rho 0.39) · GPA rated least important (3.21/5) |
| Events per parameter | 29.2 (minimum 10) |

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| Learning curve | Validation C 0.641 → 0.693 from 25% → 100% of training data, levelling off |
| Usability | SUS 76.3 (95% CI 69.5–83.0), 10 of 12 ≥ 68, alpha 0.90 · benchmark 68 |
| Would use it | 64.3% of graduates · 70.3% of employers rated it useful |
| Fairness (test) | C: women 0.724, men 0.574 (intervals overlap) |
| Software | 295 automated tests pass · code public on GitHub |

## 3. Why this research

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| **1. Why would I use it?** | It is the only tool that gives a Sri Lankan IT graduate a realistic *time range* to a first IT job, explains what helps or holds them back, and compares their skills with real employer ads. |
| **2. Will it increase my chance of a job?** | Not by itself. It shows which profiles found work sooner, so you can see what to work on, such as internships or the skills employers advertise. It is decision support, not a promise. |
| **3 / 4. Evidence it works?** | On 132 graduates the model never saw, it ranked who would find work first correctly about 62 times in 100 (C 0.619), better than ChatGPT (0.539). Among 885 graduates, higher GPA and internship experience went with clearly shorter searches. |
| **5. Can you guarantee a job?** | No. The study is observational, so it shows associations, not causes, and hiring depends on things I could not measure, like contacts and interviews. I say this in the app and the thesis. |
| **6 / 9. Why was it needed / the problem?** | About 17,000 IT graduates a year, a skills mismatch, and no individual evidence of how long the search takes. National statistics give unemployment rates, not individual timelines. |
| **7. How likely is it to work?** | It ranks moderately well. Its probabilities are too pessimistic, so the app leads with the range and labels the probabilities as indicative. |
| **8. What is the gap?** | No study combines time-to-employment survival modelling, employer and job-ad data, individual explanations and a test against a general LLM, for Sri Lankan IT graduates. |
| **10. Validation method?** | A fixed 70/15/15 split. Tuning with cross-validation on the training set only. The model was chosen on the validation set by a rule fixed in advance. The test set was used once and never re-tuned. |
| **11. How does it work?** | The graduate enters a profile → the API → the saved survival model gives a survival curve → most likely time, range and confidence → explanations, what-ifs and skill gaps against real ads. |

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| **12. Competitors?** | Rukais et al. (2025), ITConnect, Scopira and commercial job sites. They predict a category or match jobs. None predicts *when*, or uses graduates still searching. |
| **13. Why not ChatGPT?** | I tested it. On the same 132 graduates it was near chance (0.539), over-optimistic, and gave different answers to the same profile on repeated runs. It was never trained on Sri Lankan graduate outcomes. |

## 4. Methodology and data

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| **14. Primary or secondary?** | Both. Primary: graduate and employer questionnaires. Secondary: 324 real job ads and published Sri Lankan statistics. |
| **15. Why 300?** | **Correct the number.** Cochran's formula gave 384, so the target was about 400. I collected 1,175 graduates; 885 had a usable outcome for modelling. |
| **16. Sampling type?** | Non-probability: purposive, convenience and snowball, because no list of all IT graduates exists. I state it as a limitation. |
| **17. Sample and population?** | Population about 17,000 IT graduates a year. Sample 1,175 (885 modelled), 245 employers, 324 ads. |
| **18 / 19. Approach, and is it yours?** | Pragmatist philosophy, deductive and quantitative. The methods are established (survival analysis, Cox/AFT, DeepSurv); what's new is combining them for this population and question. |
| **20. Proof the data were used?** | The data cleaning log, data card, the split counts, the walkthrough notebook that re-runs every number, and 295 tests on GitHub. |
| **21. How is it evaluated?** | Model: Uno's C, integrated Brier score, calibration, fairness, robustness. App: SUS with 12 users, and the ChatGPT benchmark. Software: 295 tests and response times. |
| **22. Questionnaire validity?** | Review by 6 experts, an evidence matrix for every item, a pilot (21 graduates, 9 employers), and KMO, Bartlett, parallel analysis, AVE and HTMT. |
| **23. Reliability?** | Cronbach's alpha with confidence intervals, and omega, for the skill scales. |

## 5. Additional questions

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| **24. Class imbalance?** | It isn't a classification problem. People still searching are *censored*, and survival models use their partial information instead of dropping them. The split keeps the same share in each part. |
| **25. Overfitting?** | Composite skill scores keep 29.2 events per parameter. Elastic-net penalty, early stopping for DeepSurv, tuning only on training and validation, and a locked test set. |

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| **26. Self-reported outcome reliable?** | It's a limitation. I asked for the month and year (not "how long"), removed impossible dates, and checked sensitivity to rounded answers. |
| **27. Employer social desirability?** | Yes, 68% of ratings were 4 or 5. So I used them to *rank* skills, not as absolute levels, and checked them against real ads. The weak match (rho 0.39) is itself a finding. |
| **28. SHAP stable on small data?** | The chosen model is AFT, explained by exact time ratios, not SHAP. SHAP is only for DeepSurv. Uncertainty comes from 200 bootstrap refits. |
| **29. Explanations understandable?** | Plain-language sentences, not SHAP plots. SUS 76.3, and I added a "How to read this result" guide after user feedback. |
| **30. Fallback if too few?** | Not needed: 1,175 graduates and 245 employers, well above the targets. |
| **31. Bias by gender or institution?** | Checked C-index and calibration by gender and institution type. Gender is never a model input. The intervals overlapped, and I report the gap honestly. |
| **32. Data protection?** | No names or contacts, consent on every response, Sri Lanka's PDPA 2022, the app stores no profile (tested), and institutions are anonymised. |
| **33. Why survival, not a classifier?** | A quarter were still searching. A classifier would drop them or mislabel them as "unemployed", and it can't say *when*. |
| **34. Causation?** | Correlation only. The thesis and the app both say so. |
| **35. One-sentence novelty** | "The first explainable survival-analysis system that predicts *when* Sri Lankan IT graduates find a first IT job, triangulated with employer and job-ad data, and tested against a general LLM." |
| **36. Why no Random Forest / XGBoost?** | Each user needs an individual explanation, and Cox/AFT give interpretable time ratios. With small data, simpler models are safer. It was also the supervisor's requirement, and a test in the code enforces it. |

## 6. The hard ones (prepare these most)

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| **"0.62 is weak."** | "It's moderate, and it's honest: a test set used once, not tuned. Contacts and interview skill weren't measured. My learning curve shows more data would add only about 0.01–0.02, and more complex models didn't beat it. That's why the app shows ranges." |
| **"No better than Kaplan–Meier on error / it under-predicts."** | "The ranking works, but the probabilities are pessimistic, by up to 17 points. The evaluation sets are small and later cohorts found jobs faster. I reported this instead of re-tuning on the test set, and the app labels probabilities as indicative." |
| **"Weak for a BSc."** | "Three data sources, seven survival models compared under a pre-set rule, a locked test, fairness and robustness checks, a SUS study, an LLM benchmark, a working app with 295 tests, and a public reproducible repository." |

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| **"Will society use it? The market is crashing."** | "64.3% of graduates said they would use it and 70.3% of employers found it useful. In a hard market, knowing your realistic timeline and gaps matters more, not less." |
| **"Hiring is human."** | "Agreed. It doesn't replace hiring. It shows which preparation patterns went with faster hiring, so a graduate can act earlier." |
| **"How is ChatGPT's model different?"** | "ChatGPT is a general language model trained on web text to predict words. It wasn't trained on Sri Lankan graduate outcomes, can't use people still searching, and isn't consistent. Mine is trained on 620 real local graduates for this one outcome. ChatGPT is better at general advice and wording." |
| **"So what does a graduate do differently?"** | "Start an internship early, look at the skill gaps employers actually advertise (cloud, testing, DevOps), and plan with a realistic timeline instead of guessing." |
| **"Non-random sample, a third from one institution."** | "Agreed, and stated. Compared with Kelaniya 2022, my sample found work faster, so the results describe this sample, not all Sri Lankan graduates." |
| **"SUS users were your peers."** | "Yes, a stated limitation. The SUS measures perceived usability, and their comments led to real changes in the app." |
| **"You changed the feature set after pre-registering."** | "It's documented as a deviation with its reason. The core and extended sets give almost the same result (difference 0.003)." |
| **"Is it just a form plus a model?"** | "No: a survival model with uncertainty, explanations, what-ifs, peer comparison, real job-ad skill gaps and a grounded chat. Every part is tested." |
| **"The pilot is in your main data."** | **Fix the thesis wording before the viva.** Answer: "The questions didn't change after the pilot, so those responses were kept in the main sample." |
| **"Your GitHub has a random forest."** | "That was a one-off check of whether a more complex model could beat mine. None did, so it is not part of the system, as the supervisor directed." |

## 7. Glossary (if they ask "what is…?")

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| **Survival analysis** | Methods for "time until an event" that also use people for whom the event hasn't happened yet. |
| **Censoring** | We know the graduate was still searching at the survey date, but not when they will find a job. |
| **Kaplan–Meier** | A curve of the share still searching over time, accounting for censoring. My baseline. |
| **Cox model** | Models the *rate* of finding a job; each factor multiplies that rate (hazard ratio above 1 = sooner), without assuming the shape of the time curve. |
| **AFT model** | Models the time itself; each factor stretches or shrinks it (time ratio). The one I selected (log-normal). |

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| **DeepSurv** | A neural-network version of the Cox model that can learn non-linear patterns. |
| **Uno's C-index** | Out of pairs of graduates, how often the model correctly says who finds work first; 0.5 = chance, 1 = perfect. Suited to censored data. |
| **Integrated Brier score** | Average error of the predicted probabilities over time; lower is better. |
| **Calibration** | Whether predicted percentages match what actually happened. |
| **SHAP** | A way to split a prediction into each factor's contribution; used for DeepSurv. |
| **Pragmatism** | Choosing whichever methods best answer the question, rather than one fixed philosophy. |
| **SUS** | A standard 10-question usability score from 0 to 100; 68 is average. |
| **Holm correction** | Adjusts p-values when many tests are run, so chance findings aren't over-claimed. |

## 8. Don't say

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| "300 responses" | Say 1,175 collected, 885 modelled. |
| "It will get you a job" / "causes" | Say "is associated with", "graduates like you". |
| "The model is accurate" | Say "it ranks moderately well; probabilities are indicative". |
| "I tuned it until it improved" | Never. The test set was used once. |